

Parallel Computing (CS 633)

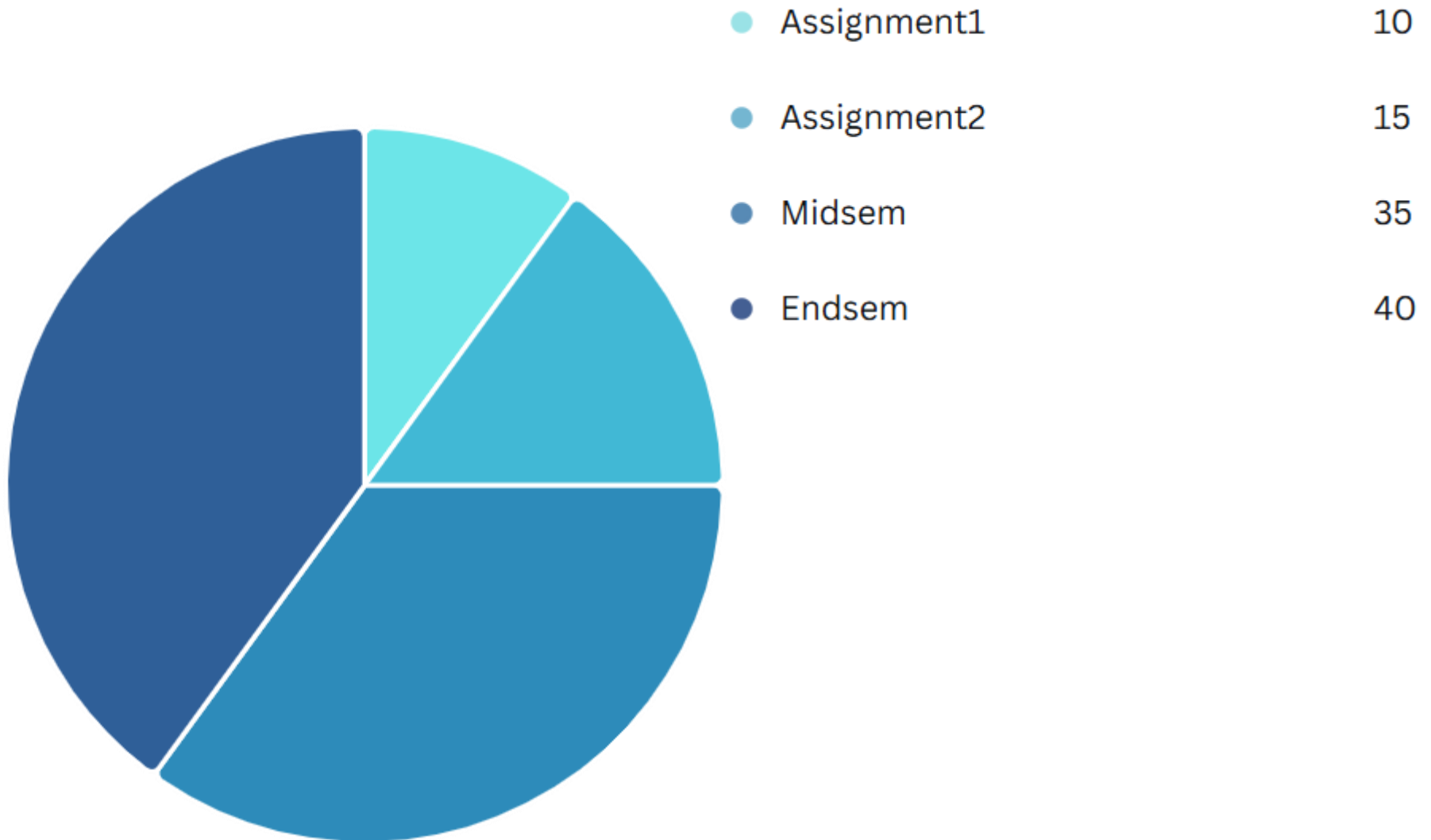
January 5, 2026

Preeti Malakar
pmalakar@cse.iitk.ac.in

Logistics

- Class hours: MW 3:30 – 5:00 PM (L7)
- Office hour: Sat 4:00 – 5:00 PM (KD 221)
- <https://www.cse.iitk.ac.in/users/cs633/2025-26-2>
 - Lectures will be uploaded after every class
- Extra class/doubts: Saturday 11:30 AM – 12:30 PM
- Announcements/uploads on
 - MookIT (hello.iitk.ac.in)
 - Course email alias

Grading Policy



Attendance

- No weightage
 - 70% attendance is compulsory for this course
- Compulsory for good grades 😊
- Attendance will be taken daily on Acadly (time will be decided by the instructor)
- **From DoAA**
 - **Attendance:** As per the Senate minutes (576th Meeting, Item #16), attendance for all lectures must be recorded, irrespective of whether it is used for grading.

Lectures

- Lecture slides are pointers for the topic
 - They won't be as verbose as a book!
- In case you miss a class, please ensure you are up to date with the lecture content
 - Either ask your most trusted friend
 - Ask the instructor

Any question? Ask!

- Email [should always be prefixed with **[CS633]** in the subject]
- After class
- Office hour (Saturdays 4 – 5 PM)
- Saturday doubt session (will be announced)
- <https://forms.gle/CuBWQgLrg1YjNNuS9>

Switch OFF All Devices



INDIAN INSTITUTE OF TECHNOLOGY KANPUR
ACADEMIC SECTION

No. DOAA/20010-11/Mobile

October 4, 2010

Use of Mobile Phones in Academic Area

The following policy will be followed regarding the usage of Mobile Phones in the Academic Area:

Examinations: Students are not permitted to carry Mobile phones inside the examination hall. The faculty members/invigilators must keep the mobile phones switched off during the conduct of the examination.

Classrooms: Mobile phones are to be switched off in class-rooms both, by students as well as Instructors/Tutors.

Laboratory/Library/Auditorium: Mobile phones are to be kept in silent mode in laboratories/library/auditorium. In case the individual would like to receive/make a call, they must do so from outside the premises.

The implementation of the above will be overseen by the following:

Examinations: Instructors/Invigilators

Classrooms: Instructors/Tutors

Laboratory: Instructors/Tutors/Officer-in-charge of the Laboratory

Library: Librarian

Auditorium: Facility-in-charge.


Sanjay Mittal

Dean of Academic Affairs

Assignment

- Two programming assignments in C using MPI
- In a group (group size = 5)
 - Send group member information by Jan 14 via Google forms (link will be shared by Jan 6)
 - Include names, roll numbers, IITK email-ids
- Mode of submission will be explained in due time

Assignments

- Two
- Penalty for late submission
- Cannot be completed in a day!
- Discussion is NOT allowed outside your group
 - You are responsible to maintain your code and report within your group only
 - You will be penalized
 - If you copied code from another group
 - If your code was copied from by another group

Plagiarism

Plagiarism will NOT be tolerated

Use of AI tools is NOT allowed

Message from CSE DUGC

1. It is imperative that you uphold the highest standards of academic honesty in all your coursework and assessments. Engaging in unfair means—such as plagiarism, cheating, or unauthorized collaboration—undermines not only your own learning but also the credibility of our institution. Any violations of academic integrity will be dealt with strictly as per institute guidelines, which may include significant penalties. Also as a department we have an Anti Cheating Policy (<https://www.cse.iitk.ac.in/pages/AntiCheatingPolicy.html>). Please make sure you understand and adhere to the policy. If you have doubts about what constitutes fair practices, please do not hesitate to ask your course instructor or myself.
2. I must also remind you that forging or tampering with official documents, including exam copies, grade reports, medical certificates or any correspondence, is a serious violation of ethical and legal standards. Such actions will not be tolerated under any circumstances and may result in disciplinary action. Please remember that maintaining trust and credibility is as important as academic performance.

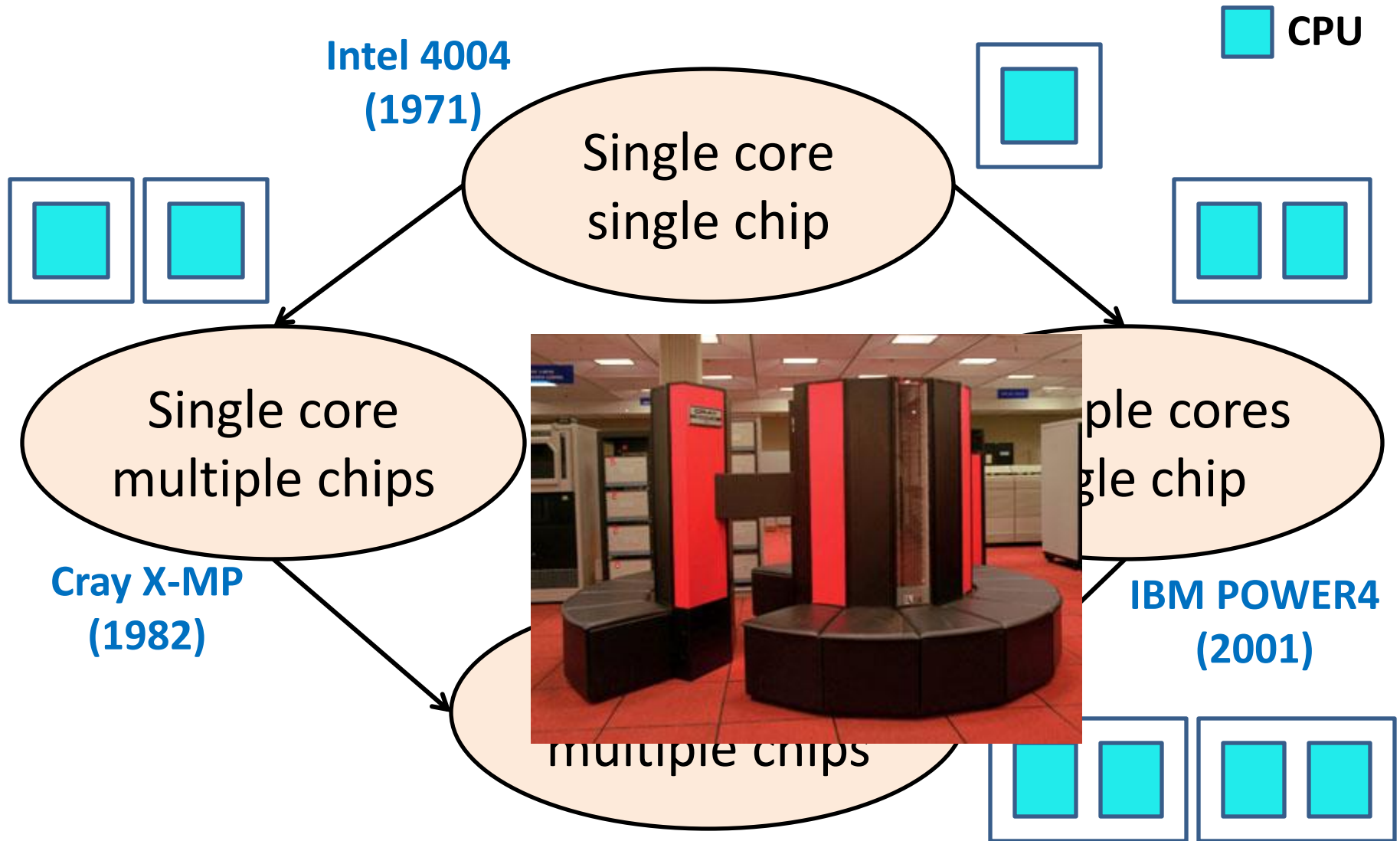
Message from CSE DUGC

3. Regular attendance in lectures and active participation are essential for your success in this course. You are expected to remain up to date with the course material from the very beginning of the semester, as concepts build progressively and gaps can quickly become difficult to bridge. Missing classes not only hampers your understanding of the subject but also disrupts the continuity of the learning process. You are therefore expected to attend all lectures regularly and stay actively engaged with the course material.
4. Lastly, I would like to emphasize that you are not alone in your academic or non-academic challenges. If you face any issues—whether related to the course, personal circumstances, or administrative concerns—please feel free to reach out to me or the department head (head@cse.iitk.ac.in). We are here to guide you, and ensure that you have the resources you need to thrive during your time at IIT Kanpur.

Lecture 1

Introduction

Multicore Era



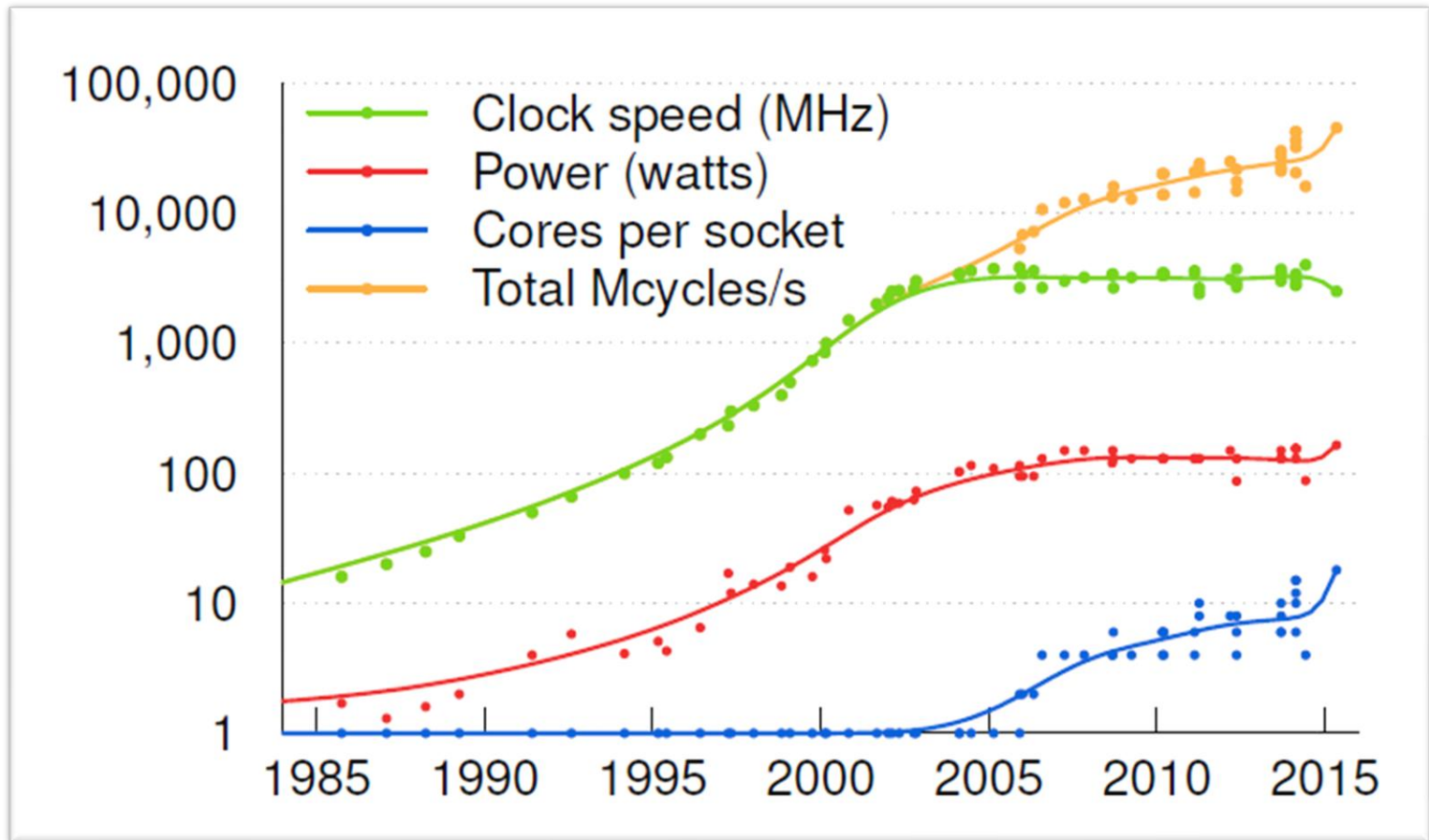
Number of transistors in a chip doubles every 18 months



[Source: Wikipedia]

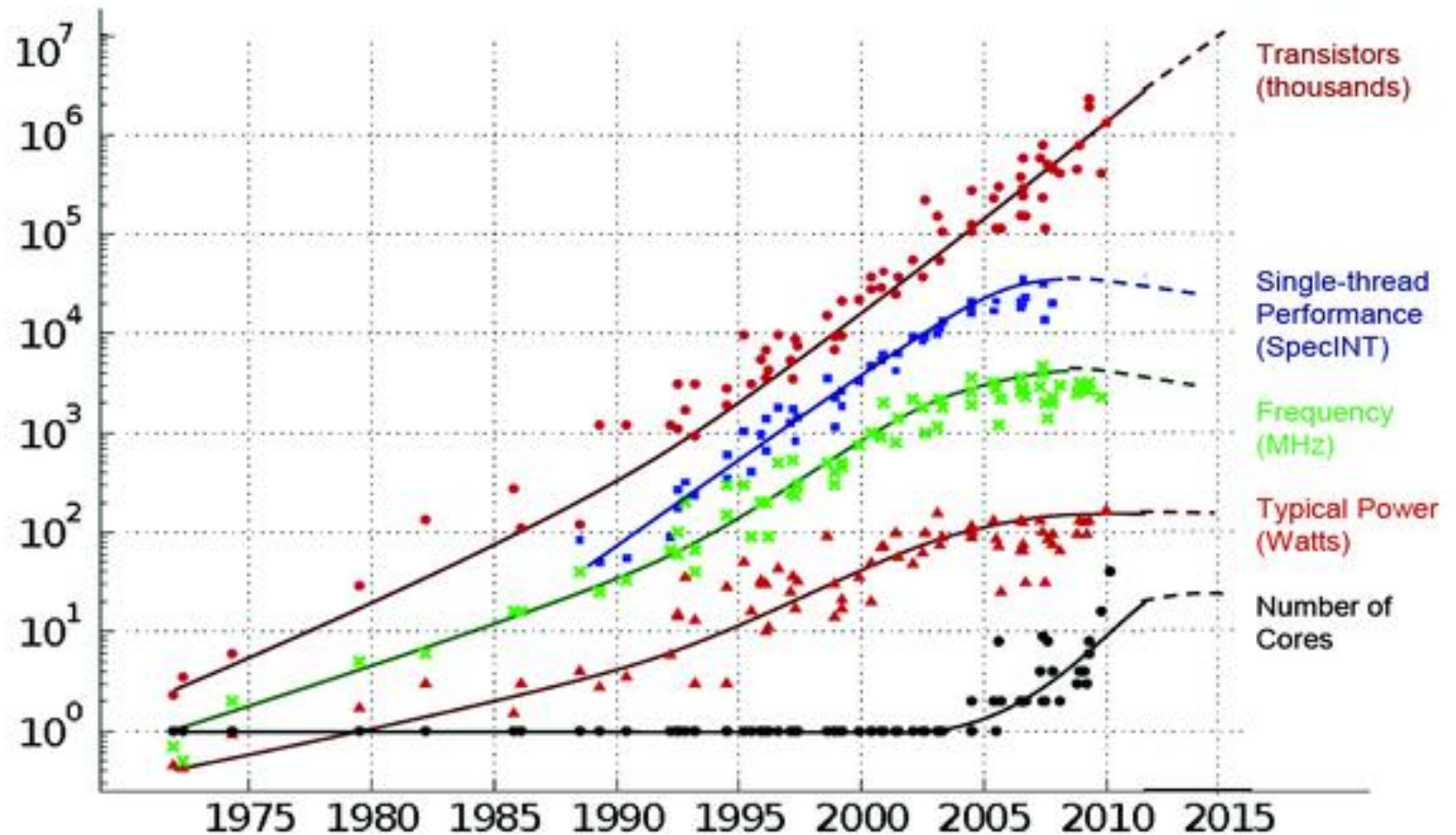
16

Trends



[Source: M. Frans Kaashoek, MIT]

35 YEARS OF MICROPROCESSOR TREND DATA



Original data collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond and C. Batten
Dotted line extrapolations by C. Moore

top500.org (Nov'24)

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory	9,066,176			
		9,264,128			
		2,073,600	561.20	846.84	



~ \$600 million
~ 7300 sq. ft.
~ 22 MW power
~ 23000 L water

2022 List: Start of Exascale Era!

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,730,112	1,102.00	1,685.65	21,100
2	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
3	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,220,288	309.10	428.70	6,016
4	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, Atos EuroHPC/CINECA Italy	1,463,616	174.70	255.75	5,610
5	Summit - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	148.60	200.79	10,096

Top 500 Supercomputers 2025 List

1 EFLOP/s = 10^{18} FLOP/s



Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	793.40	930.00	13,088
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	

green500.org (Nov'23)

Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	Power (kW)	Energy Efficiency (GFlops/watts)
1	293	Henri - ThinkSystem SR670 V2, Intel Xeon Platinum 8362 32C 2.8GHz, NVIDIA H100 80GB PCIe, Infiniband HDR, Lenovo Flatiron Institute United States	8,288	2.88	44	65.396
2	44	Frontier TDS - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	120,832	19.20	309	62.684
3	17	Adastra - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Grand Equipement National de Calcul Intensif - Centre Informatique National de l'Enseignement Suprieur (GENCI-CINES) France	319,072	46.10	921	58.021

Metric of interest: Performance per Watt


[HOME](#)
[COMPLETE RESULTS](#)
[GREEN GRAPH500](#)
[SUBMISSIONS](#)
[BE](#)

RANK ↕	PREVIOUS RANK ↕	MACHINE ↕	VENDOR ↕	TYPE ↕	NETWORK ↕	INSTALLATION SITE ↕	LOCATION ↕
1	new	Supercomputer Fugaku	Fujitsu	Fujitsu A64FX	Tofu Interconnect D	RIKEN Center for Computational Science (R-CCS)	Kobe Hyogo
2	new	Wuhan Supercomputer	HUST	Kunpeng 920+Tesla A100	Custom	Wuhan Supercomputing Center	Wuhan
3	3	Frontier	HPE	HPE Cray EX235a	Slingshot-11	DOE/SC/Oak Ridge National Laboratory	Oak Ridge TN
4	new	Pengcheng Cloudbrain-II	HUST-Pengcheng Lab-HUAWEI	Kunpeng 920+Ascend 910	Custom	Pengcheng Lab	ShenZhen

Top #1 Supercomputer

<https://www.top500.org/resources/top-systems/>



El Capitan: DOE/LLNL
No.1 in Nov 2024



Frontier: DOE/SC/Oak Ridge National
Laboratory
No.1 in Jun 2022



Supercomputer Fugaku: RIKEN Center for
Computational Science
No.1 from Jun 2020 until Jun 2022



Summit: DOE/SC/Oak Ridge National
Laboratory
No.1 from Jun 2018 until Nov 2019



Sunway TaihuLight: National
Supercomputing Center in Wuxi
No.1 from Jun 2016 until Nov 2017



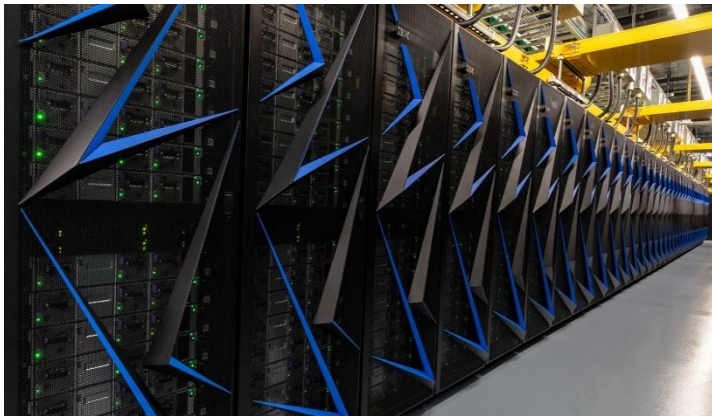
Tianhe-2 (MilkyWay-2) : National University
of Defense Technology
No.1 from Jun 2013 until Nov 2015

HPL-MXP MIXED-PRECISION BENCHMARK

The HPL-MxP benchmark seeks to highlight the emerging convergence of high-performance computing (HPC) and artificial intelligence (AI) workloads. While traditional HPC focused on simulation runs for modeling phenomena in physics, chemistry, biology, and so on, the mathematical models that drive these computations require, for the most part, 64-bit accuracy. On the other hand, the machine learning methods that fuel advances in AI achieve desired results at 32-bit and even lower floating-point precision formats. This lesser demand for accuracy fueled a resurgence of interest in new hardware platforms that deliver a mix of unprecedented performance levels and energy savings to achieve the classification and recognition fidelity afforded by higher-accuracy formats.

<https://hpl-mxp.org/>

Making of a Supercomputer



Source: energy.gov

Greenest Data Centre?



Source: MIT TR 06/19

“The 149,000 square foot facility built on a hillside overlooking the UC Berkeley campus and San Francisco Bay will house one of the most energy-efficient computing centers anywhere, tapping into the region’s mild climate to cool the supercomputers at the National Energy Research Scientific Computing Center (NERSC) and eliminating the need for mechanical cooling.”

BERKELEY LAB OPENS STATE-OF-THE-ART FACILITY FOR COMPUTATIONAL SCIENCE

Wang Hall takes advantage of Lab’s hillside location for advanced energy efficiency

NOVEMBER 12, 2015

Contact: Jon Bashor, jbashor@lbl.gov, 510-486-5849

A new center for advancing computational science and networking at research institutions and universities across the country opened today at the Department of Energy’s (DOE) Lawrence Berkeley National Laboratory (Berkeley Lab).

Named Shyh Wang Hall, the facility will house the National Energy Research Scientific Computing Center, or NERSC, one of the world’s leading supercomputing centers for open science which serves nearly 6,000 researchers in the U.S. and abroad. Wang Hall will also be the center of operations for DOE’s Energy Sciences Network, or ESnet, the fastest network dedicated to science, connecting tens of thousands of scientists as they collaborate on solving some of the world’s biggest scientific challenges.

Complementing NERSC and ESnet in the facility will be research programs in applied mathematics and computer science that develop new methods for advancing scientific discovery. Researchers from UC Berkeley will also share space in Wang Hall as they collaborate with Berkeley Lab staff on computer science programs.



Top Supercomputers from India (Nov'23)

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
90	AIRAWAT - PSAI - NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHz, NVIDIA A100, Infiniband HDR, Netweb Technologies Center for Development of Advanced Computing (C-DAC) India	81,344	8.50	13.17	
163	PARAM Siddhi-AI - NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHz, NVIDIA A100, Mellanox HDR Infiniband, EVIDEN Center for Development of Advanced Computing (C-DAC) India	41,664	4.62	5.27	
201	Pratyush - Cray XC40, Xeon E5-2695v4 18C 2.1GHz, Aries interconnect , HPE Indian Institute of Tropical Meteorology India	119,232	3.76	4.01	1,353
354	Mihir - Cray XC40, Xeon E5-2695v4 18C 2.1GHz, Aries interconnect , HPE National Centre for Medium Range Weather Forecasting India	83,592	2.57	2.81	955

2024...

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
136	AIRAWAT - PSAT - NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHz, NVIDIA A100, Infiniband HDR, Netweb Technologies Center for Development of Advanced Computing (C-DAC) India	81,344	8.50	13.17	
188	Arka - BullSequana XH2000, AMD EPYC 7643 48C 2.3GHz, Infiniband HDR, Red Hat Enterprise Linux, EVIDEN Indian Institute of Tropical Meteorology India	290,016	5.94	7.40	1,236
189	Arunika - BullSequana XH2000, AMD EPYC 7643 48C 2.3GHz, Infiniband HDR, Red Hat Enterprise Linux, EVIDEN National Centre for Medium Range Weather Forecasting India	203,040	5.94	7.40	1,236
268	Pratyush - Cray XC40, Xeon E5-2695v4 18C 2.1GHz, Aries interconnect , HPE Indian Institute of Tropical Meteorology India	119,232	3.76	4.01	1,353
400	Arka AI/ML - NVIDIA DGX H100, Xeon Platinum 8480C 56C 2GHz, NVIDIA H100 SXM5 80GB, Octo-rail NVIDIA HDR100 Infiniband, Red Hat Enterprise Linux, EVIDEN Indian Institute of Tropical Meteorology India	8,176	2.70	3.75	
431	Mihir - Cray XC40, Xeon E5-2695v4 18C 2.1GHz, Aries interconnect , HPE National Centre for Medium Range Weather Forecasting India	83,592	2.57	2.81	955

Supercomputing in India [topsc.cdacb.in, Jul'24]

Rank	Site	System	Core/Processor/Socket/Nodes	Rmax (TFlops)	Rpeak (TFlops)
1	Center for Development of Advanced Computing (C-DAC),PUNE	AIRAWAT - PSAI is a NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHZ, NVIDIA A100, INFINIBAND HDR2 More Info OEM:NVIDIA, Bidder:Netweb	81344/2/82	8500	13176
2	Indian Institute of Tropical Meteorology(IITM),Pune	Cray XC-40 class system with 3315 CPU-only (Intel Xeon Broadwell E5-2695 v4 CPU) nodes with Cray Linux environment as OS, and connected by Cray Aries interconnect. Total storage More Info OEM:Cray, Bidder:Cray	119232//3315	3763.9	4006.19
3	National Centre for Medium Range Weather Forecasting (NCMRWF),Noida	Intel Xeon Broadwell E5-2695 v4 CPU) nodes with Cray Linux environment as OS, and connected by Cray Aries interconnect More Info OEM:Cray, Bidder:Cray	83592/2/2612	2570.4	2808.7
4	PARAM Pravega,IISc,Bangalore	The PARAM Pravega is a heterogeneous and hybrid configuration of Intel Xeon Cascade Lake processors,NVIDIA Tesla V100 with NVLink, Mellanox HDR More Info OEM:ATOS, Bidder:ATOS	29952/2/624	1702	2565
5	Indian Institute of Technology (IITK),Kharagpur	The supercomputer PARAM Shakti is based on a heterogeneous and hybrid configuration of Intel Xeon Skylake(6148, 20C, 2.4Ghz) processors, and NVIDIA Tesla V100. The system More Info OEM:Atos, Bidder:Atos	17280/2/442	935	1290.2

Supercomputing in India [topsc.cdacb.in, Dec'25]

Rank	Site	System	Core/Processor/Socket/Nodes	Rmax (TFlops)	Rpeak (TFlops)
1	Center for Development of Advanced Computing (C-DAC),PUNE	AIRAWAT - PSAI is a NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHZ, NVIDIA A100, INFINIBAND HDR2 More Info OEM:NVIDIA, Bidder:Netweb	81344/2/82	8500	13176
2 	Indian Institute of Tropical Meteorology(IITM),Pune	Arka - BullSequana XH2000, AMD EPYC 7643 48C 2.3GHz, Infiniband HDR, Red Hat Enterprise Linux 2 More Info OEM:EVIDEN, Bidder:EVIDEN	290016/2/3021	8280.72	10610.8
3 	National Centre for Medium Range Weather Forecasting (NCMRWF),Noida,Noida	Arunika - BullSequana XH2000, AMD EPYC 7643 48C 2.3GHz, Infiniband HDR, Red Hat Enterprise Linux, 2115 Nodes and 203,040 cores2 More Info OEM:EVIDEN, Bidder:EVIDEN	203040/2/2115	5960	7400
4	Indian Institute of Tropical Meteorology(IITM),Pune	Cray XC-40 class system with 3315 CPU-only (Intel Xeon Broadwell E5-2695 v4 CPU) nodes with Cray Linux environment as OS, and connected by Cray Aries interconnect. Total storage More Info OEM:Cray, Bidder:Cray	119232//3315	3763.9	4006.19
5 	Indian Institute of Tropical Meteorology(IITM),Pune	Arka AI/ML - BullSequana XH2000, AMD EPYC 7643 48C 2.3GHz, Infiniband HDR, Red Hat Enterprise Linux56 More Info OEM:EVIDEN, Bidder:EVIDEN	8176/2/7	2700	3750

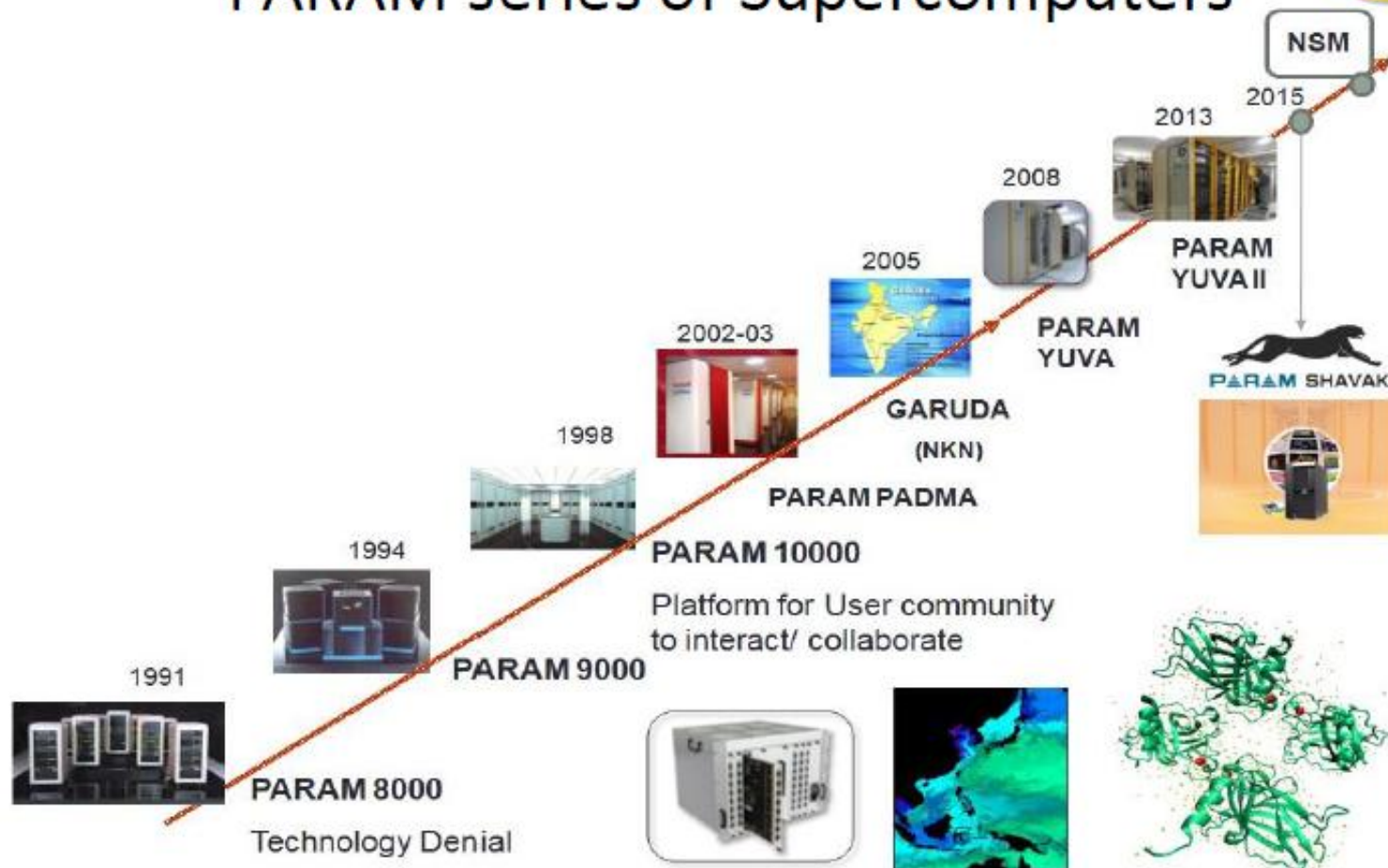


Param Sanganak
New HPC facility being installed @IITK

Source: www.iitk.ac.in

Rank	Site	System	Core/Processor/Socket/Nodes	Rmax (TFlops)	Rpeak (TFlops)
11	Sankhya Sutra Labs Ltd,Bangalore	AMD Rome based Cluster256-Node Compute,AMD Rome based Cluster, AMD EPYC 7742 64-Core Processor with 128core,2 Master nodes,2 login nodes and Mellanox HDR 100 More Info OEM:HPE, Bidder:Cray	32768/2/256	922.48	1179.65
12	National Atmospheric Research Laboratory(NARL),Tirupati	The Supercomputer PARAM AMBAR is based on a heterogeneous and hybrid configuration of Intel Xeon Cascade Lake processors, NVIDIA Tesla V100 with nvlink, 100Gbps Intel OPA 100% More Info OEM:Tyrone, Bidder:Netweb	18816//	919.61	1384.85
13	Supercomputer Education and Research Centre (SERC), Indian Institute of Science (IISc),Bangalore	The Cray XC-40 System installed at SERC, IISc.(1468 Intel Xeon E5-2680 v3 @ 2.5 GHz dual twelve-core processor CPU-only nodes, 48 [Intel Xeon E5-2695v2 @ 2.4 Ghz single twelve-core More Info OEM:Cray, Bidder:Cray	36336 + 2880 + 126720//	901.51	1244
14	Indian Institute of Technology(IIT),Kanpur	The PARAM SANGANAK is a heterogeneous and hybrid configuration of Intel Xeon Cascade Lake processors, NVIDIA Tesla V100 with NVLink, Mellanox HDR More Info OEM:ATOS, Bidder:ATOS	13834//292	851.3	1350
15	Indian Institute of Tropical Meteorology,Pune	Aaditya HPC is built on IBM P6 575 nodes totaling 117 numbers including 4 GPFS nodes and 2 Login nodes. Each node is populated with 32 cores of IBM P 6 CPU running at 4.7 G Hz.The total More Info OEM:IBM, Bidder:IBM	38016/16/117	719.2	790.7

PARAM series of Supercomputers



Credit: Ashish Kuvelkar, CDAC

National Supercomputing Mission Sites

Completed NSM Sites

PARAM Shivay



PARAM Shakti



PARAM Brahma



PARAM Yukti



PARAM Sanganak



PARAM Pravega



PARAM Seva



PARAM Smriti



PARAM Utkarsh

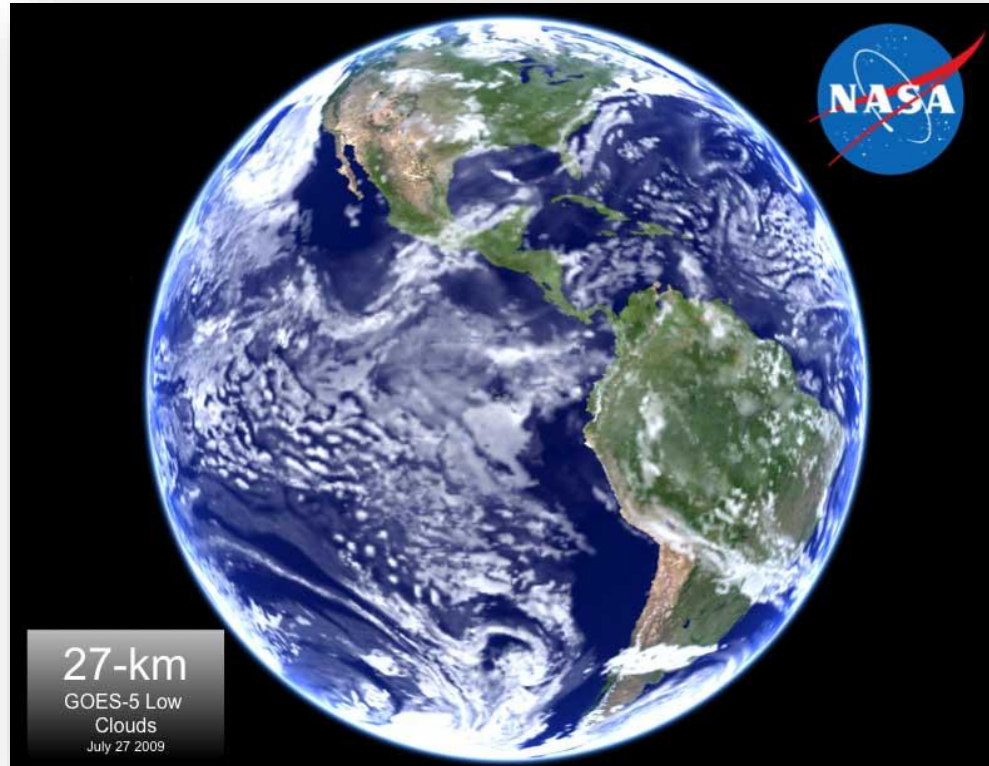


PARAM Ganga



Big Compute

Massively Parallel Codes



Climate simulation of Earth [Credit: NASA]

Discretization

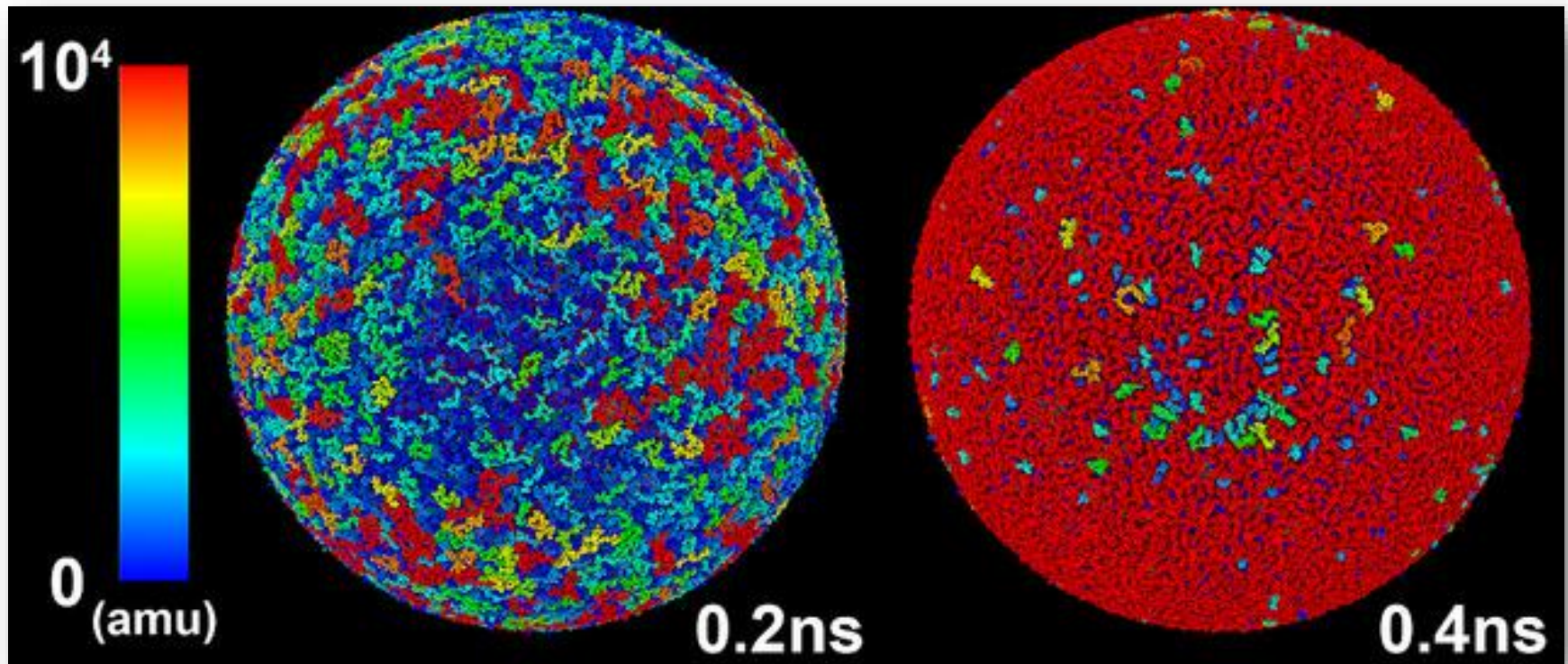


Gridded mesh for a global model [Credit: Tompkins, ICTP]

Numerical Weather Models

- Use numerical methods to solve equations that govern atmospheric processes
- Are based on fluid dynamics and depend on observations of meteorological variables
- Are used to obtain nowcast/forecast

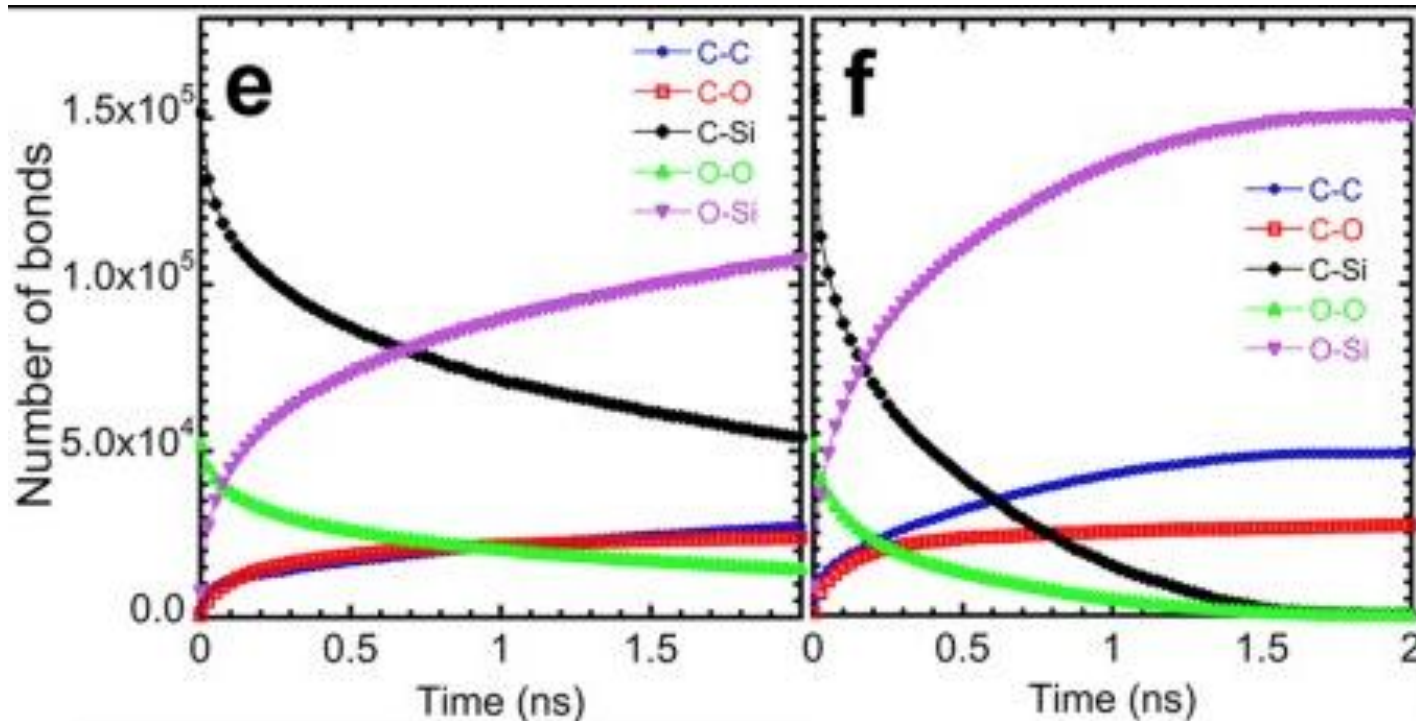
Massively Parallel Simulations



Self-healing material simulation

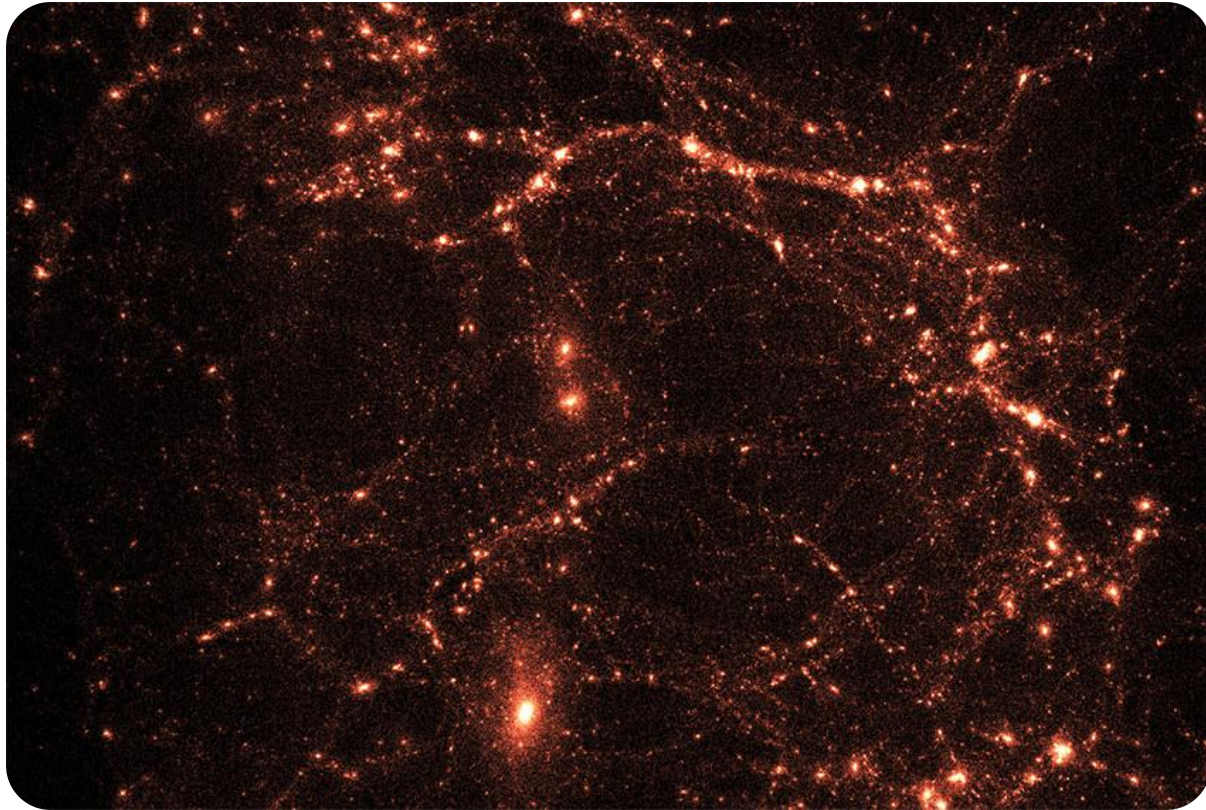
[Nomura et al., “[Nanocarbon synthesis by high-temperature oxidation of nanoparticles](#)”, Scientific Reports, 2016]

Massively Parallel Analysis



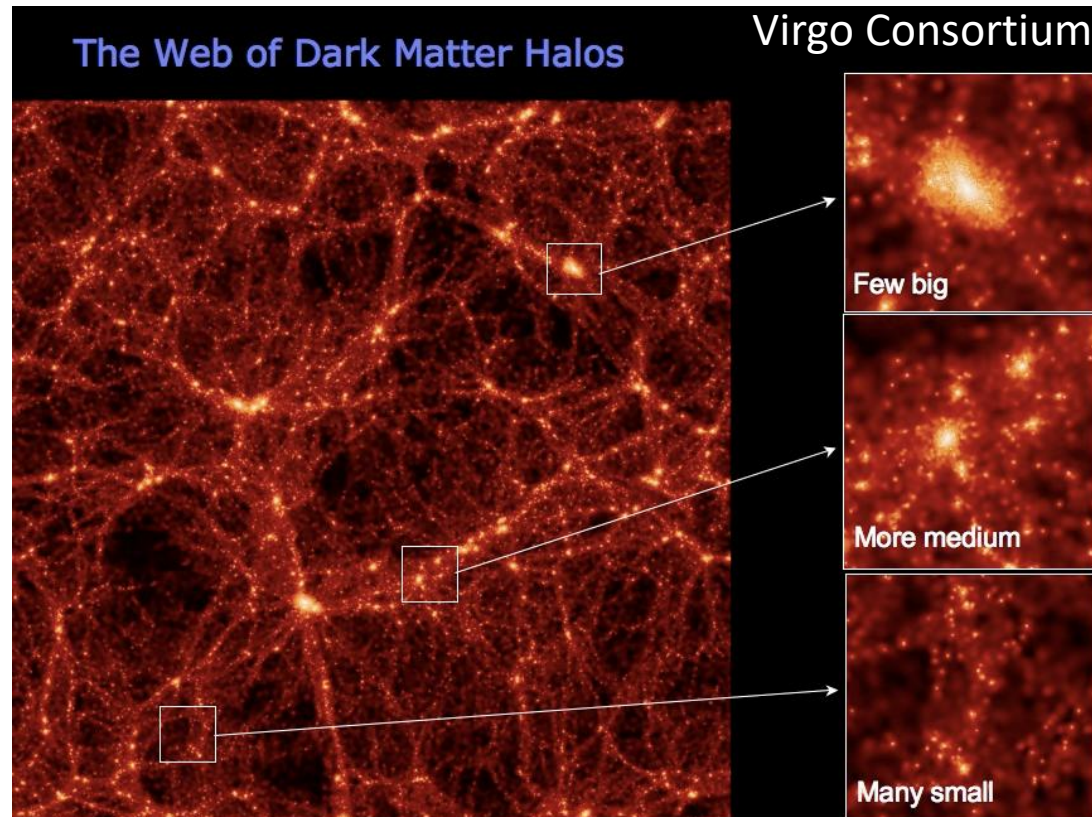
[Nomura et al., “[Nanocarbon synthesis by high-temperature oxidation of nanoparticles](#)”, Scientific Reports, 2016]

Massively Parallel Codes

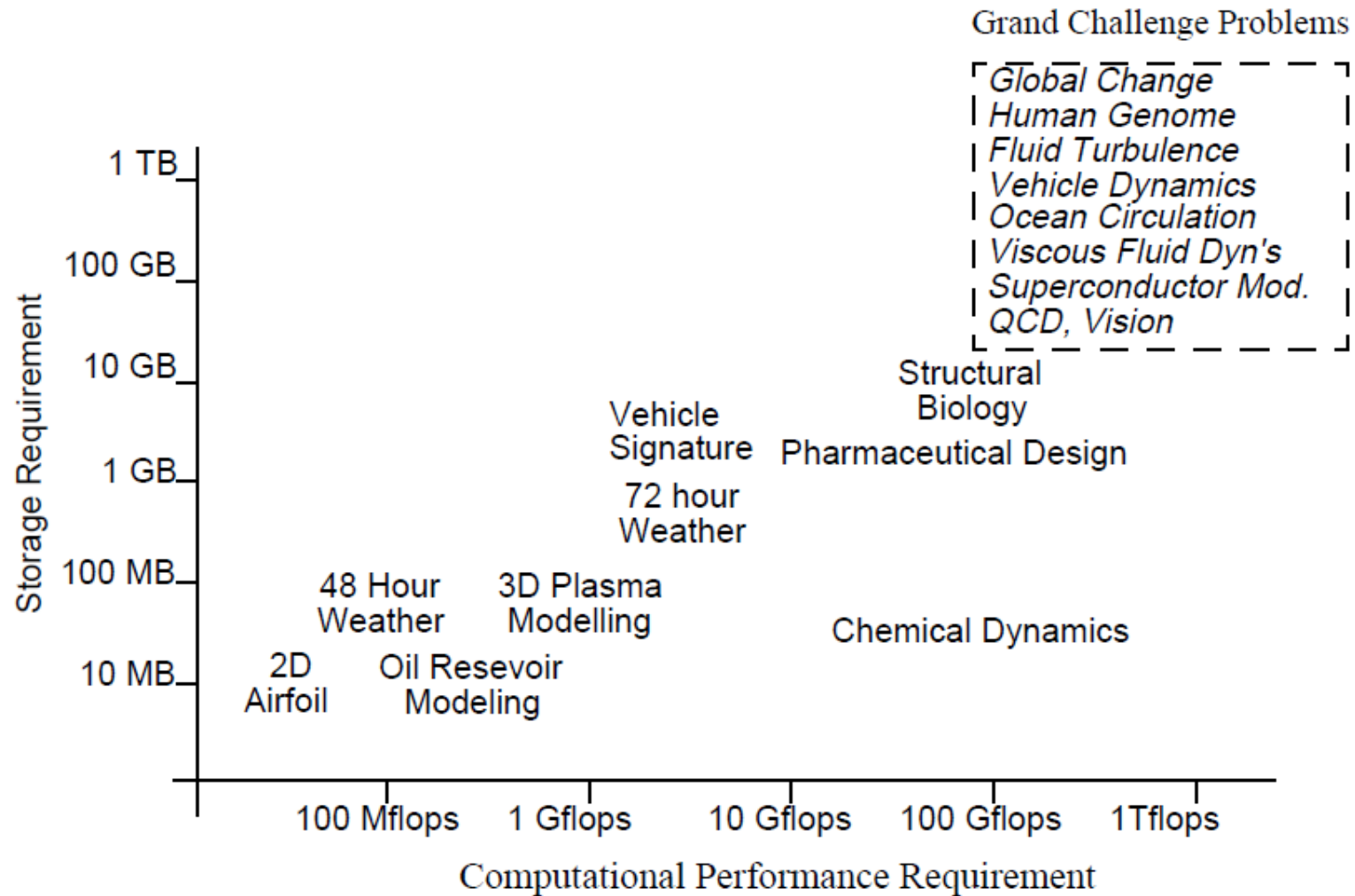


Cosmological simulation [Credit: ANL]

Massively Parallel Analysis



Computational Science

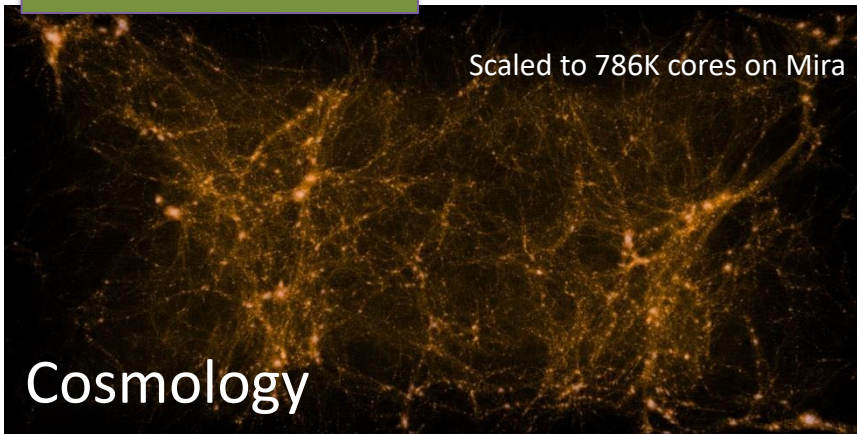


[Source: Culler, Singh and Gupta]

Big Data

Output Data

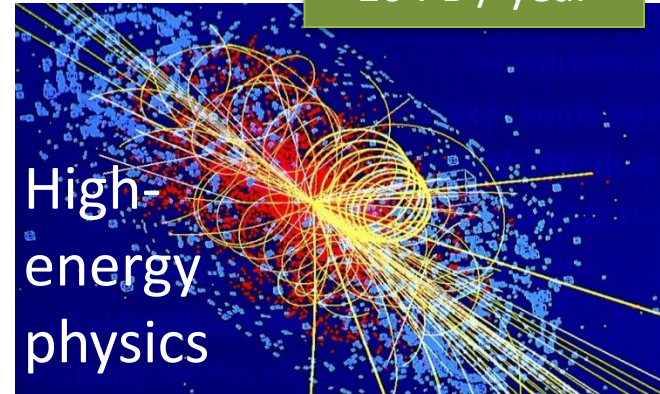
2 PB / simulation



Scaled to 786K cores on Mira

Q Continuum simulation
Source: Salman Habib et al.

10 PB / year



Higgs boson simulation

Source: CERN

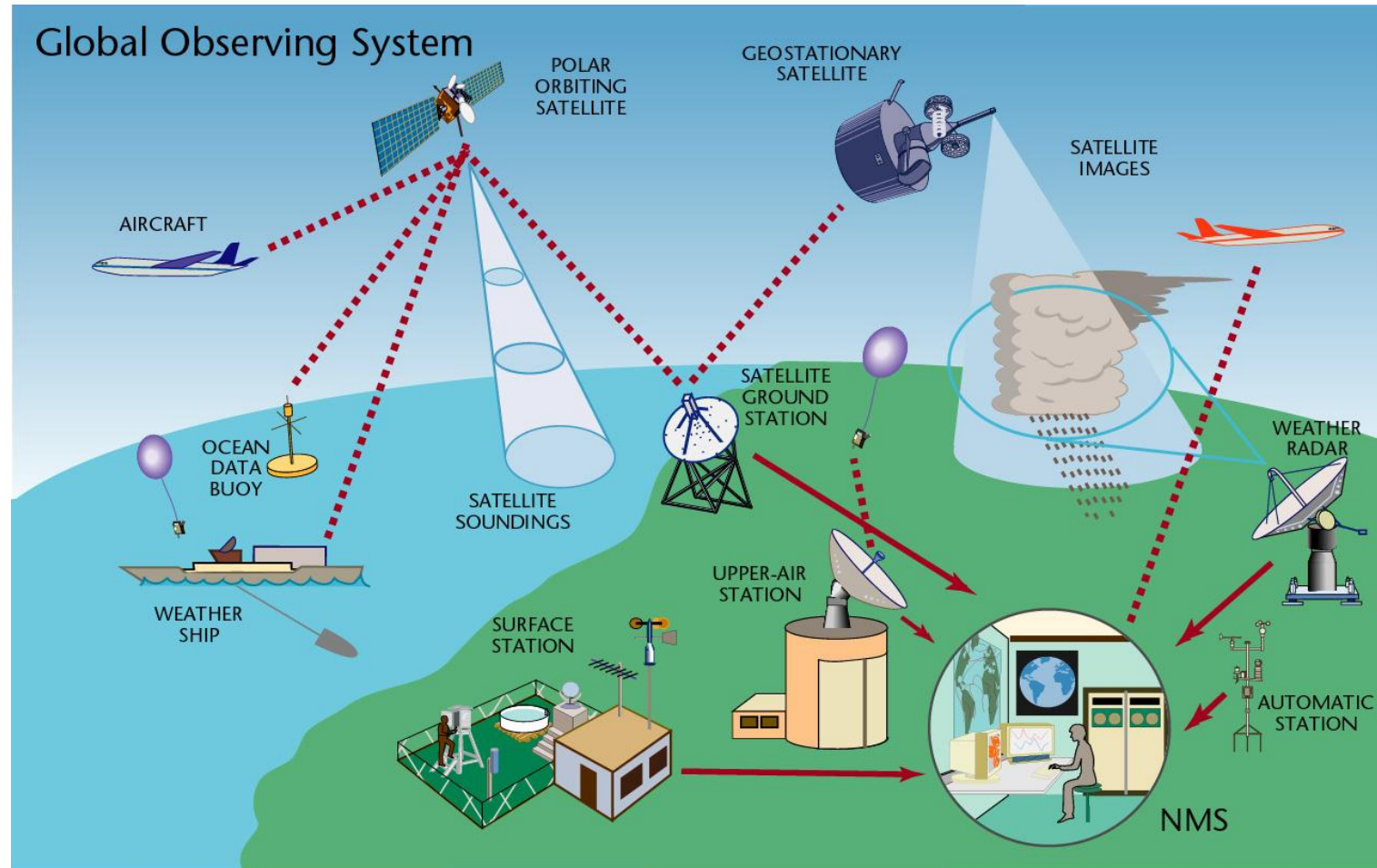
240 TB / simulation



Hurricane simulation

Source: NASA

Input Data



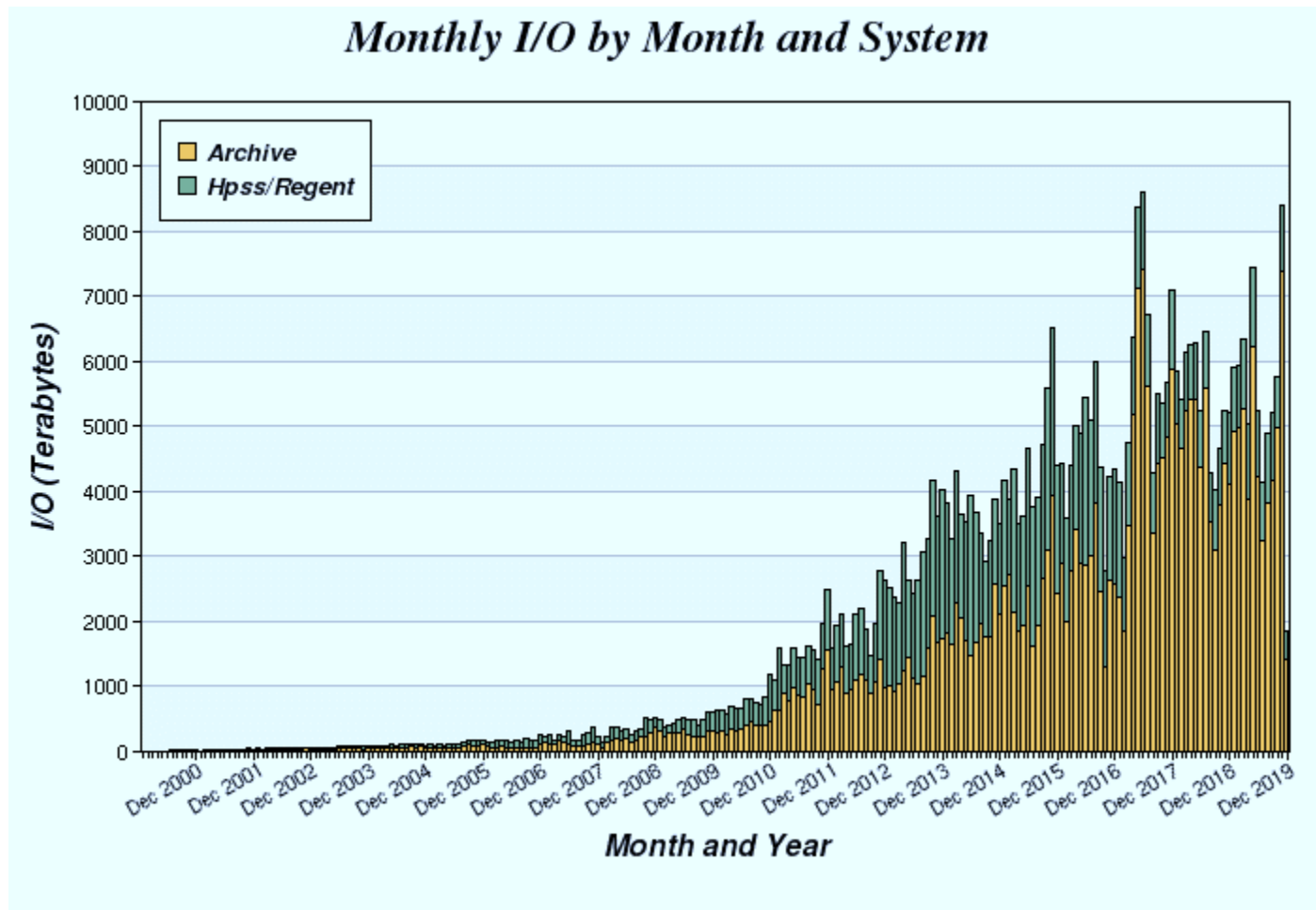
[Credit: World Meteorological Organization]

System Architecture Trends

System attributes	2010	2017-2018		2021-2022	
System peak	2 Peta	150-200 Petaflop/sec		1 Exaflop/sec	
Power	6 MW	15 MW		20 MW	
System memory	0.3 PB	5 PB		32-64 PB	
Node performance	125 GF	3 TF	30 TF	10 TF	100 TF
Node memory BW	25 GB/s	0.1TB/sec	1 TB/sec	0.4TB/sec	4 TB/sec
Node concurrency	12	O(100)	O(1,000)	O(1,000)	O(10,000)
System size (nodes)	18,700	50,000	5,000	100,000	10,000
Total Node Interconnect BW	1.5 GB/s	20 GB/sec		200GB/sec	
MTTI	days	O(1day)		O(1 day)	

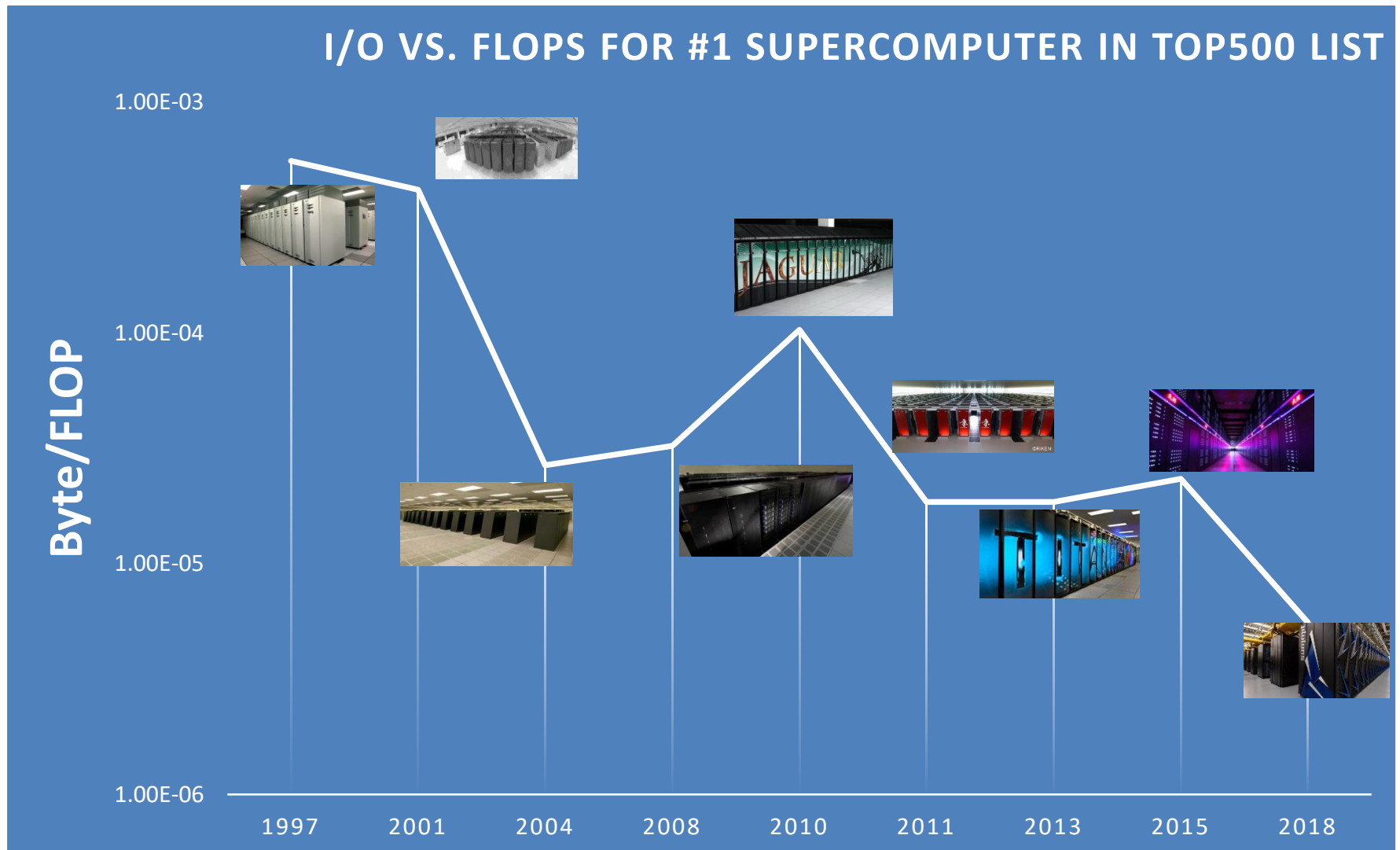
[Credit: Pavan Balaji@ATPESC'17]

I/O trends

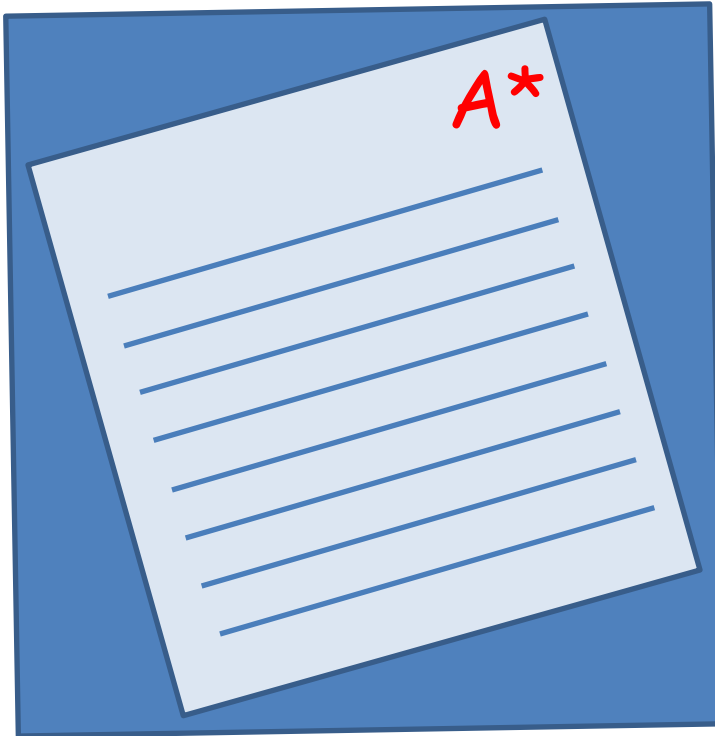


NERSC I/O trends [Credit: www.nersc.gov]

Compute vs. I/O trends



Why Parallel?



20 hours



2 hours

Not really

Parallelism

A parallel computer is a collection of processing elements that communicate and **cooperate** to solve large problems **fast**.

– Almasi and Gottlieb (1989)

Speedup

Example – Sum of squares of N numbers

Serial

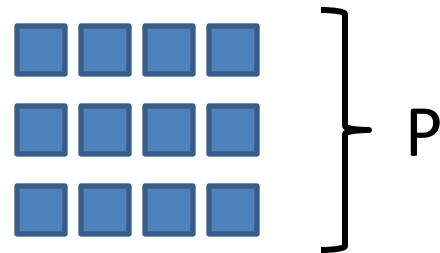
```
for i = 1 to N  
  sum += a[i] * a[i]
```



$O(N)$

Parallel

```
for i = 1 to N/P  
  sum += a[i] * a[i]  
collate result
```



$O(N/P) +$

Communication time

Performance Measure

- Speedup

$$S_p = \frac{\text{Time (1 processor)}}{\text{Time (P processors)}}$$

- Efficiency

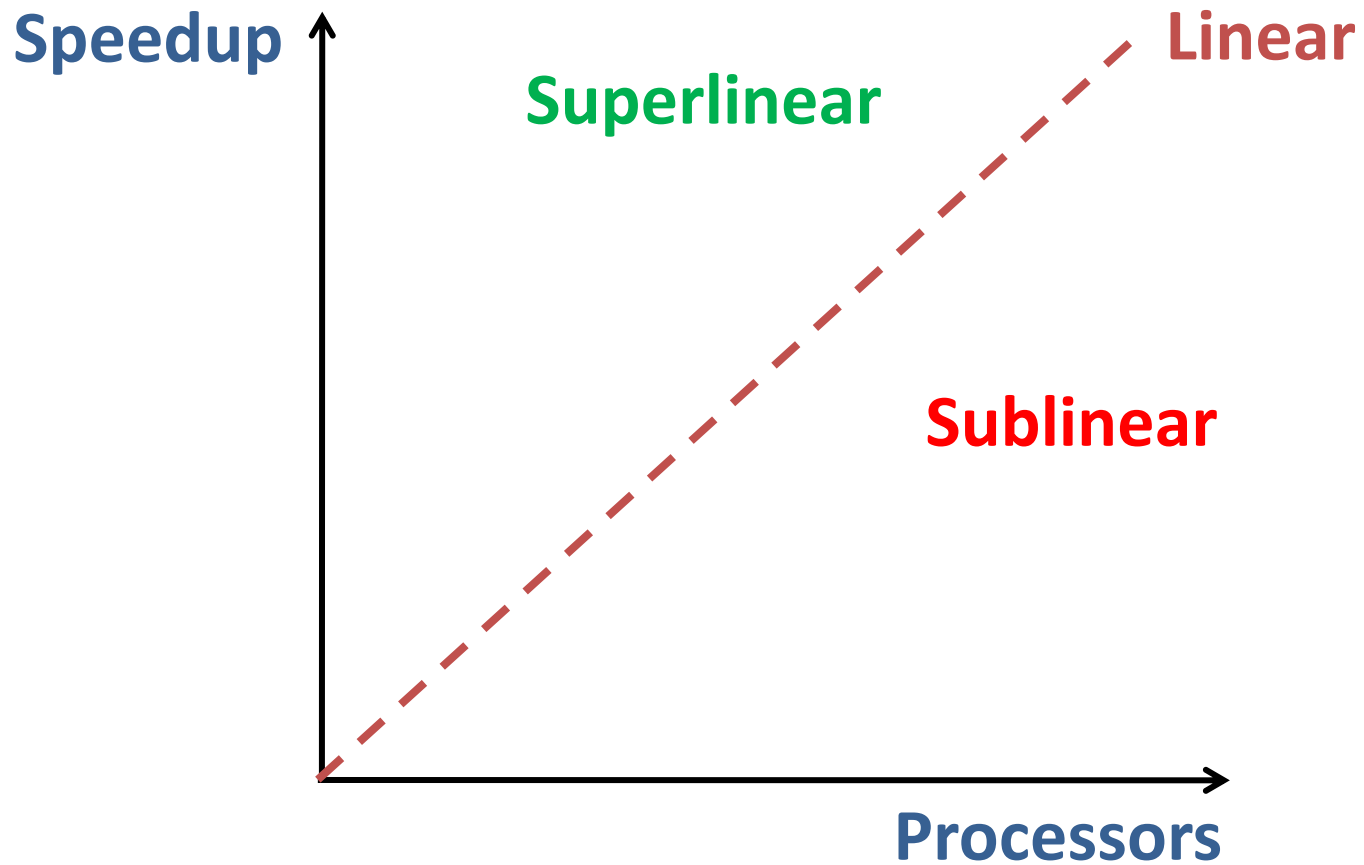
$$E_p = \frac{S_p}{P}$$

Parallel Performance (Parallel Sum)

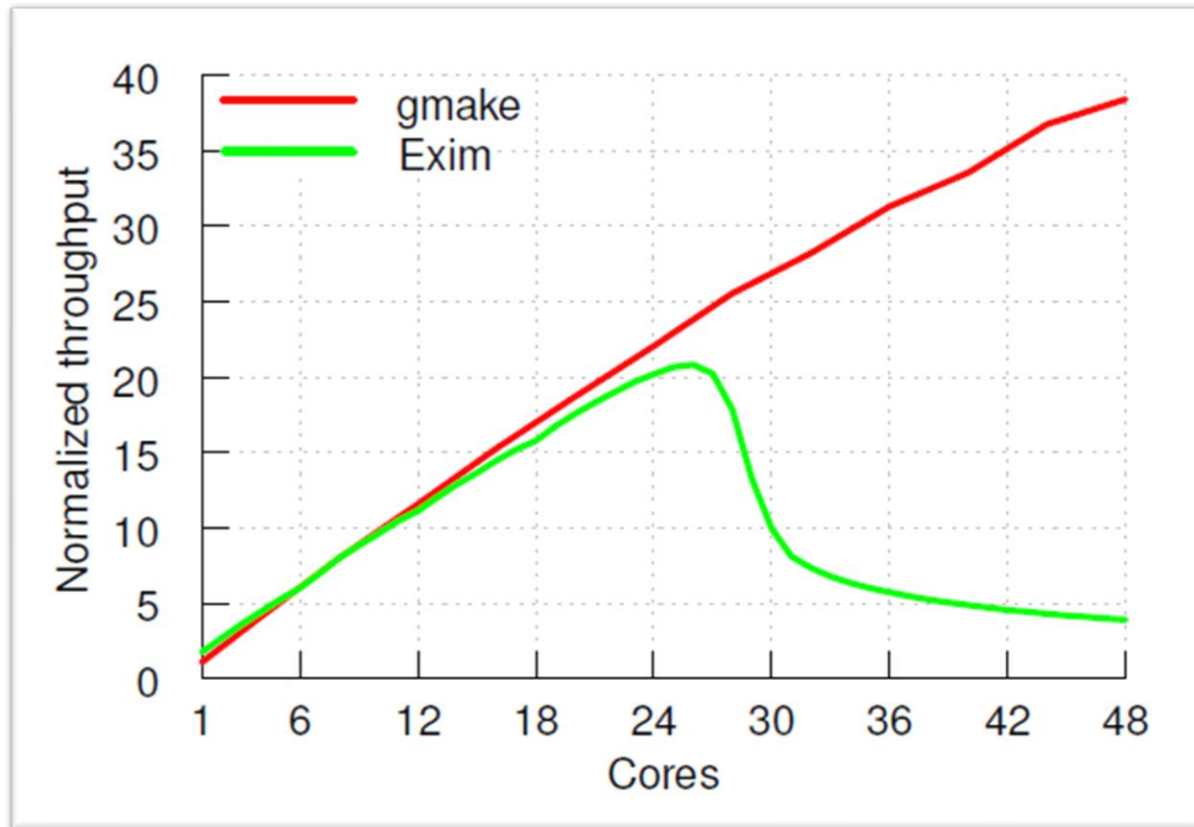
Parallel efficiency of summing 10^7 doubles

#Processes	Time (sec)	Speedup	Efficiency
1	0.025	1	1.00
2	0.013	1.9	0.95
4	0.010	2.5	0.63
8	0.009	2.8	0.35
12	0.007	3.6	0.30

Ideal Speedup

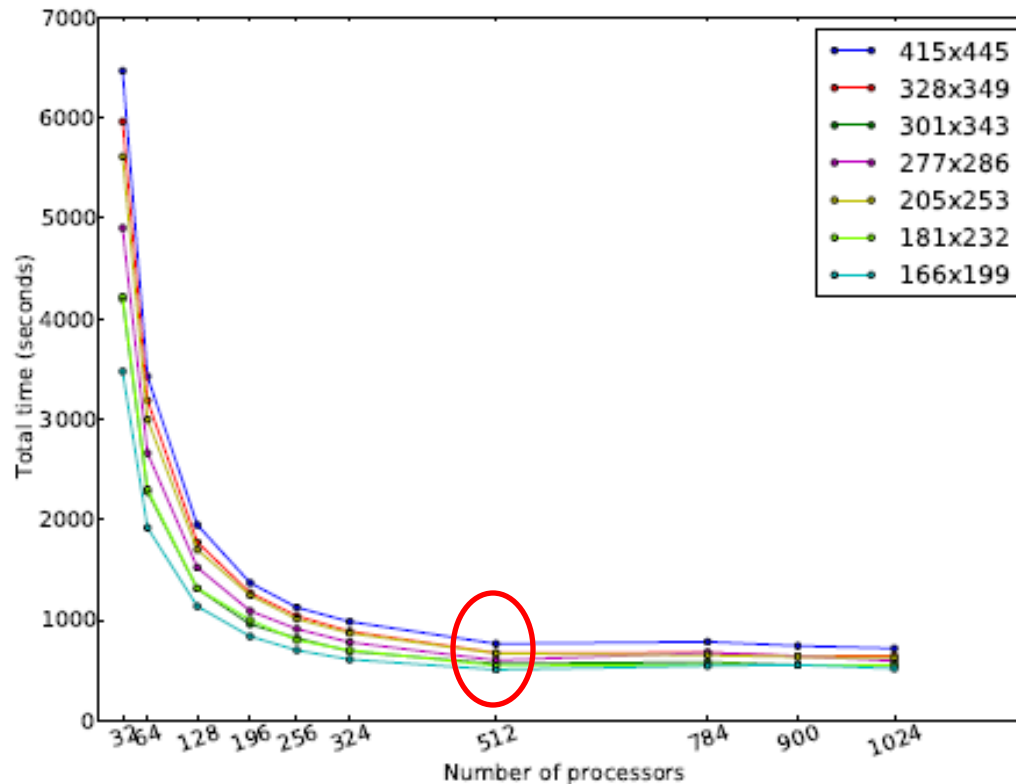


Issue – Scalability



[Source: M. Frans Kaashoek, MIT]

Scalability Bottleneck



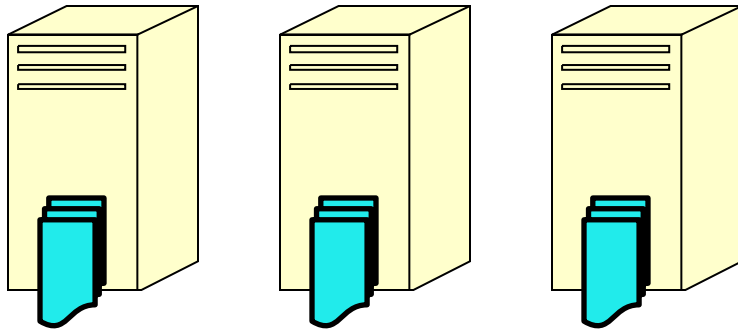
Performance of weather simulation application

Parallelism

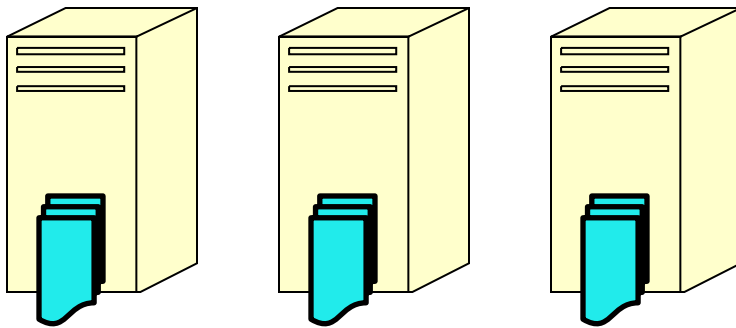
A parallel computer is a collection of processing elements that **communicate** and cooperate to solve large problems fast.

– Almasi and Gottlieb (1989)

Distributed Memory Systems



Node



Cluster

- Networked systems
- Distributed memory
 - Local memory
 - Remote memory
- Parallel file system

Parallel Programming Models

Libraries	MPI, TBB, Pthread, OpenMP, ...
New languages	Haskell, X10, Chapel, ...
Extensions	Coarray Fortran, UPC, Cilk, OpenCL, ...

- Shared memory
 - OpenMP, Pthreads, ...
- Distributed memory
 - MPI, UPC, ...
- Hybrid
 - MPI + OpenMP

This course ...

Large-scale Parallel Computing

Message
passing

Parallel
algorithms

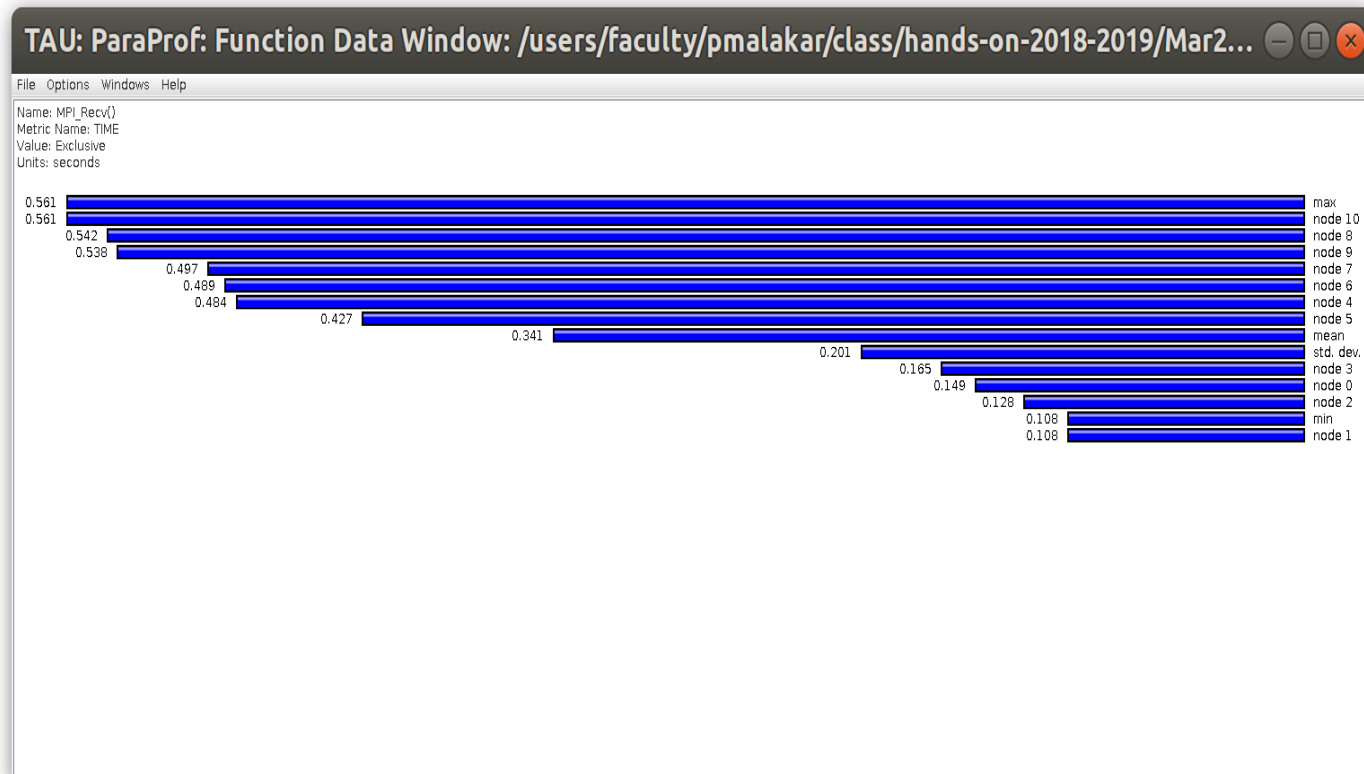
Designing
parallel codes

Performance
analysis

Message Passing Paradigm

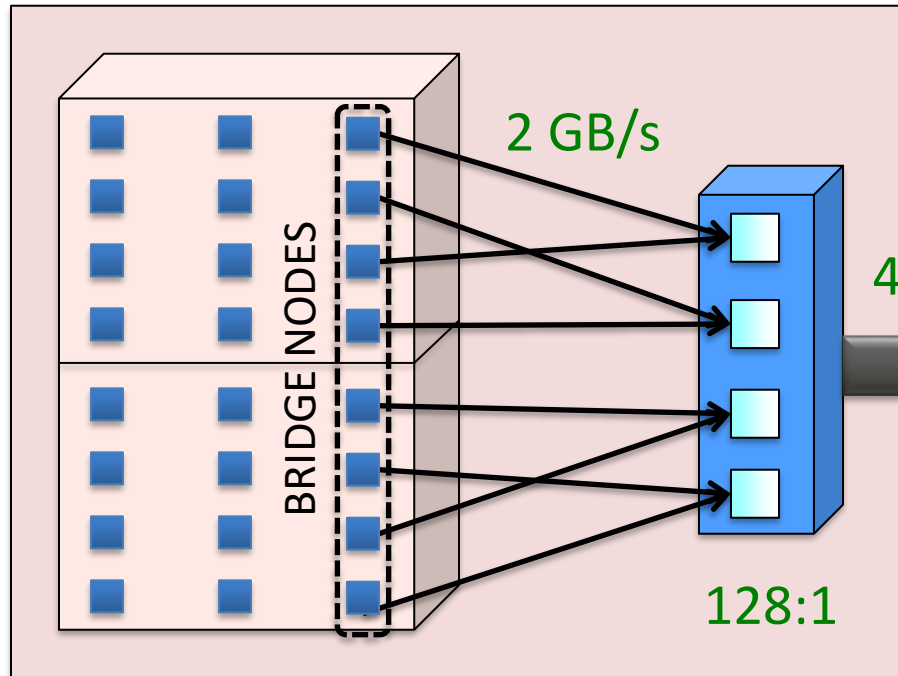
- Point-to-point (P2P) communications
- Collective communications
- Algorithms
- Performance

Profiling



Parallel I/O

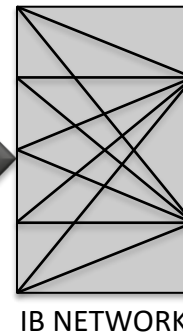
NOT SHARED



Compute node rack

I/O nodes

SHARED



IB NETWORK

GPFS filesystem

Job Scheduling



NODES



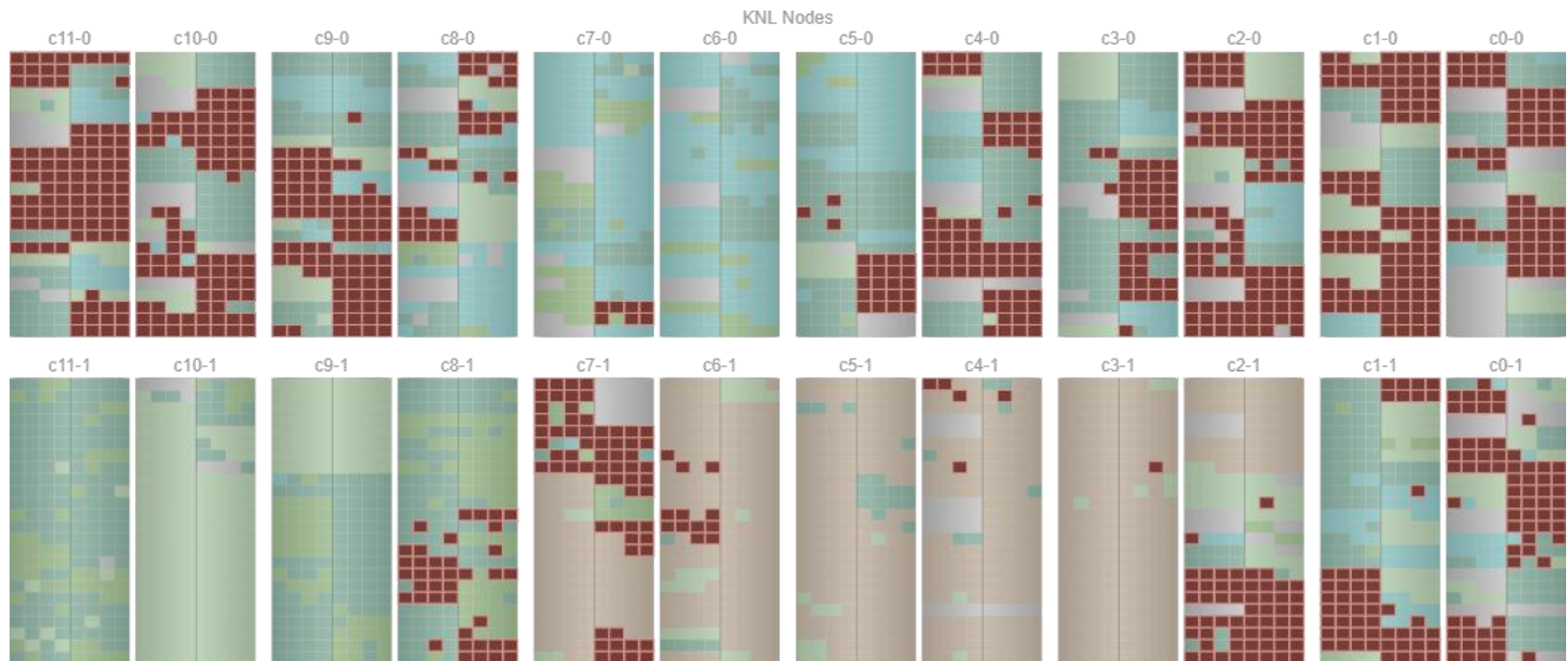
JOBs



USERS

Example of a real supercomputer activity
- [Argonne National Laboratory Theta jobs](#)

Supercomputer Activity



Reference Material

- DE Culler, A Gupta and JP Singh, Parallel Computer Architecture: A Hardware/Software Approach Morgan-Kaufmann, 1998.
- A Grama, A Gupta, G Karypis, and V Kumar, Introduction to Parallel Computing. 2nd Ed., Addison-Wesley, 2003.
- Marc Snir, Steve W. Otto, Steven Huss-Lederman, David W. Walker and Jack Dongarra, MPI - The Complete Reference, Second Edition, Volume 1, The MPI Core.
- Bill Gropp, Using MPI, Third Edition, The MIT Press, 2014.
- Research papers